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The impact of tokenization on the trading process costs and carbon emission: Empirical study on the ODDO BHF Bond

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Abstract

Emergence of blockchain technology has disrupted a number of economic sectors, particularly financial institutions, with significant effects on their operations. This paper investigates the impact of asset tokenization on the issuance and trading process of financial assets, specifically bonds. It examines the effect of tokenizing the High Yield Bond on the Ethereum blockchain across two key dimensions: On costs, a comparative cost–benefit analysis is conducted before and after tokenization, and on green sustainability, through a comparative analysis on the carbon footprint of the bond before and after Ethereum's merge to proof of stake. The results show that Tokenization improves cost-savings, and it promotes a greener, more sustainable approach when using the Ethereum blockchain post-transition to proof of stake.

Keywords Blockchain, Tokenization, Smart contract, Cost analysis, Comparative analysis, Carbon footprint, The merge, Smart bond

Introduction

The finance industry is currently struggling with a trust crisis, primarily due to past crises and scandals, such as the 2008 financial crisis, the Enron scandal, and the Lehman Brothers collapse. These events have significantly weakened trust between financial institutions and clients. In response, regulatory authorities have been actively working on implementing new risk management measures, including various adjustments to the Basel framework [1] and by introducing other frameworks, like Enterprise Risk Management (ERM), which are designed to mitigate different types of risks such as credit risk, operational risk, and market risk.

To further secure transactions and prevent cybersecurity breaches, the finance industry is also focusing on enhancing cybersecurity measures. Important cybersecurity incidents, such as the Bangladesh Bank heist in 2016 and the Equifax data breach in 2017 [2] showed the importance of having robust cybersecurity protocols of financial systems in order to maintain their security, integrity and trustworthiness.

Meanwhile, emergence of blockchain technology in 2008 introduced the concept of disintermediation, posing initially a threat for traditional financial institutions to their current offerings and positioning. Nevertheless, as research progressed, the financial sector started reconsidering its standpoint on blockchain technology and began to turn this threat into an opportunity by investigating more about the potential benefits that this technology could have, to enhance the trading process efficiency and ensure security and transparency of banking services. Blockchain is now considered as a way

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offering significant potential for enhancing risk management in finance [3, 4].

In light of this, alliance between experts in finance and blockchain has helped to expand the capabilities of this technology, assisting the introduction of a range of smart financial products that use advanced blockchain features notably stablecoins, tokenization, and smart contracts.

As a new technology, assessment of the impact of blockchain is still ongoing. While several studies demonstrated its positive impact across a wide range of industries, blockchain presents a lot of uncertainty. Additionally, regulators are conducting thorough research into the feasibility of integrating blockchain into financial institutions to prevent future crises. That is why investing in it requires determination and well-considered choices that are supported by robust risk management strategies.

Blockchain provides the infrastructure for tokenization, offering a decentralized ledger that enables recording transactions through a computer's network. Once data is recorded, it cannot be modified or deleted, offering a reliable and tamper-proof record of transactions. Transparency is fundamental in blockchain, all transactions are visible in the network, which promotes trust and reduces the risk of fraud [5]. This decentralization ensures that no single entity has control over the entire blockchain, thereby enhancing security and reducing the risk of fraud [6].

Blockchains represent practical and viable solutions for reducing transaction costs in financial markets [7, 8], thus developing a body of literature around the design of efficient blockchain-based financial markets [8–10].

Although there is abundant literature on this subject, research on bond markets remains limited despite their widespread use in contemporary financial practice. In fact, the literature on bond markets generally focuses either on highly specialized applications such as carbon emission markets [7], or on general theoretical perspectives [9, 11, 12], without addressing the issue of designing blockchain-based corporate bond markets that are efficient in terms of transaction costs [8, 13]. This is problematic because transaction cost theory provides a well-established general framework for market mechanisms, offering theoretical guidance for the design of efficient blockchain-based markets. Accordingly, this work uses transaction cost theory to explain the impact of blockchain on reducing costs.

Indeed, several recent studies have shown that blockchain functions as a decentralized and secure ledger that facilitates peer-to-peer transactions without intermediaries, thereby improving data integrity, equitable resource distribution and automation reliability.

One of the key features of asset tokenization is the facilitation of automation through smart contracts. The latter enable the automation of administrative procedures

and financial transactions, which reduce manual intervention and intermediaries (like banks, accountants, and lawyers). This automation rationalizes operations and enhances security by removing counterparty risk, resulting in more efficient and cost-effective transfer processes.

According to Perleach (2019), application of smart contracts also reduces paperwork and the complexities associated with securities management. This disintermediation could enable investment banks alone to achieve savings of \$12 billion per year and stimulate project development by reducing administrative costs (Duchenne, 2018).

Payment of dividends from assets can also be automated and transaction costs reduced, enabling regular micro-dividend and cross-border dividend payments to investors (WEF 2016; Varma, 2019).

Given the important role played by blockchain technology in accelerating transactions and creating a more efficient and accessible market, discussions about this technology have moved on to the renewable energy markets. Blockchain technology automates energy transactions, enhances transparency and reduces transaction costs, and currently the adoption of tokenization in the renewable energy sector is gradually accelerating. Bonds and Environment-Social-Governance (ESG) investments are attracting growing interest and are driving initiatives and regulatory changes around the world to align with sustainability goals. Nevertheless, blockchain is seen as a double-edged sword, because on the one hand it generates consistent energy consumption with a considerable carbon footprint, while on the other it reduces transaction costs and improves efficiency.

Furthermore, to justify the role that blockchain plays in environmental sustainability, we draw on institutional theory and stakeholder theory, two theories that reveal multi-actor forces and dynamics and provide insight into how blockchain enables, limits, or reshapes sustainability outcomes in practice.

According to institutional theory, companies adopt blockchain for efficiency reasons, but also to respond to growing stakeholder demands for real-time transparency and verifiable ESG data [14] and [15]). Freeman, the founder of stakeholder theory, argues that companies must balance the interests of all stakeholders, not just shareholders [16]. Blockchain's open data, smart contracts and immutable ledger offer stakeholders great visibility into provenance, working conditions, and environmental impacts [17] and [18]). Institutional forces create the need for verifiable ESG evidence, while stakeholder engagement determines the design and success of Blockchain solutions. These theories form a basic theoretical framework to analyze catalysts, constraints, and enabling ecosystems that shape inclusive, scalable, and resilient sustainable transitions through blockchain.

Given the importance of transaction costs associated with bond markets and the growing development of blockchain-based solutions to reduce these costs, this study aims to fill the related gaps by contributing to the literature on the impact of blockchain technology. Specifically, it raises and answers the following research question: **“Could tokenization be an effective means to increase cost effectiveness for bonds and maintain environmental sustainability?”**

Moreover, this study aims to explore the impact of smart contracts and tokenization, on costs of the trading process and on its carbon footprint. Our study will assess the impact of asset tokenization on one of ODDO BHF's financial assets: The Euro High Yield Bond DP-EUR.

ODDO BHF operates in an institutional environment where financial players (banks, asset managers, regulators, investors) are facing increasing regulatory pressure on their ESG practices. In this regard, tokenization (particularly of green or ESG assets) is becoming a means of complying with new standards of traceability, auditability, and environmental transparency. To preserve its legitimacy, ODDO BHF is encouraged to turn to blockchain and tokenization to enhance its ESG transparency.

From a stakeholder perspective, tokenization makes it possible to offer traceable assets whose ESG impact can be monitored in real time, as well as differentiated ESG products that meet the requirements of responsible investors. It also facilitates automated reporting, thereby increasing the reliability and speed of ESG information communicated to stakeholders. By meeting these expectations, ODDO BHF will be able to strengthen the trust of its stakeholders, including institutional clients, ESG investors, NGOs, regulators, and even employees, who are increasingly demanding transparency on the environmental impacts of financial products. It also attracts investors concerned about sustainability and will be able to meet growing social and regulatory expectations.

The immutable blockchain ledger allows to reduce the risk of fraud and boost investor trust by ensuring that all transactions are securely recorded and easily verifiable, [19]. This helps to increase transparency and strengthens trust between stakeholders and facilitates compliance with regulatory requirements [20].

Blockchain technology offers a wide range of benefits and challenges that can be turned into opportunities for businesses. This study will focus on investigating two significant effects of blockchain technology: its influence on trade costs and its effect on environmental sustainability.

The rest of the paper is structured as follows: an overview of existing studies that focus on cost analysis and carbon emissions related to tokenization is discussed in Sect. “Literature Review”. The third section presents an empirical study on the impact of tokenization for ODDO

BHF HYB, pointing to previous literature and the specific characteristics of the bond. The last section concludes.

Literature review

Examining previous research, we focused on studies that have examined the cost efficiency of tokenization in trading processes, as well as methodologies for assessing the carbon footprint of tokenizing financial assets. Our findings revealed that, to date, no empirical study has jointly analyzed these two dimensions. Accordingly, this section provides an overview of existing research, first focusing on studies addressing cost analysis, and then those exploring the impact of tokenization on carbon emissions. Together, these studies help contextualize our contribution, as this paper aims to bridge a gap by integrating both dimensions within a single analytical framework based on smart contract gas consumption.

Tokenization cost–benefit

Traditional financial instruments reached their limits in raising the necessary capital, ensuring a stable and regular flow of investment, and managing financial risks. This compromises their ability to meet the growing demand for investment in several areas, particularly in the field of renewable energy [21]. Tokenization is a solution that enables the creation of new financial instruments to stimulate fundraising and engage small investors through smaller shares, represented as tokens, which could facilitate reaching a larger number of investors nationwide [22].

Tokenization also promotes financial inclusion [19] by making investment opportunities accessible to small investors and those excluded from the traditional financial system. It promotes broader participation by democratizing access and lowering barriers to entry and attracts a diverse range of investors [20, 23]. Tokenization offers innovative financial mechanisms, enables fractional ownership, improves liquidity, increases transparency, and reduces costs through automation. All of this helps to reduce the cost of capital and promote financial inclusion.

These advantages over the traditional financial system help to speed up settlement times and reduce costs thanks to a simpler but more flexible financial market infrastructure [24].

In order to study tokenization from a revenue-cost perspective and bearing on the assumption that in an effective crypto-currency system, users have no incentive to double spend, [25, 26] develop a general equilibrium model of a crypto-currency to measure the welfare costs of using a crypto-currency as a payment instrument. Their results showed that it is preferable to use the revenues from currency creation rather than the transaction fees to finance the costly mining process. The authors

estimate that bitcoin generates a significant welfare loss, highly greater than the welfare loss in a monetary economy. Their study estimates that Bitcoin's current design results in a significant welfare loss of 1.4% of consumption, which could be reduced to 0.08% through optimal design changes. The authors suggest that cryptocurrencies have the potential to compete with retail payment systems if scaling issues are resolved.

Tian et al. [27] point out that an asset class, such as modern REITs, USA Internet stocks, or high-yield bonds, usually takes decades to rise above 1% of the global GDP after its issuance, unlike the blockchain tokens market, which hit 0.8% of the global GDP in only two years. Cisar et al. [28] highlight the ability of blockchain technology to reduce transaction costs in bond markets by developing a bond prototype using the Ethereum blockchain protocol.

Particularly for bonds, blockchain technology has consistently shown cost savings for asset tokenization in multiple studies. In the following, we will highlight the findings of previous research on the impact of tokenization on three specific types of bonds: Islamic bonds, green bonds, and classic corporate bonds.

Khan et al. [29] experiments a cost benefit-analysis before and after tokenization of a sukuk bond in the Islamic finance industry. The conducted comparative study is founded on estimating the costs of Al-Murabaha smart sukuk, for different types of blockchain (public and consortium) by coding the basic functions of its smart contract. Referring to the gas consumption estimated, gas fees are calculated and compared to expenses of the AL-ALDAR conventional sukuk process, which are gathered from ALDAR's 2019 statements and the World Bank's bond issuance cost estimates.

The authors here focused on coding functions of the sukuk issuance process, such as "registerObligor", "new-Investor", "investInSukuk" functions and coupons "automaticPayment" function. However, they exclude Know Your Customer (KYC) verification, balance requirements, conversion from fiat to cryptocurrency function, security and privacy measures, etc. from the code. The code is written in "Solidity" language. In addition, it is deployed and tested on Remix programming tool that can be used, as well as a gas consumption estimator.

The results of this study indicated that tokenizing the studied sukuk has significantly incurred less expense compared to conventional sukuk issuance. Furthermore, the analysis shows that using private or consortium blockchains is more cost-efficient than issuing a token sukuk on a public one. While the findings rely on certain implementation factors, such as the location preference of nodes (Paris, Dubai, and Malaysia) and the quantity of investors (five), it highlights the financial benefits of blockchain-based sukuk issuance and suggests further

investigation to optimize this digital transformation for wider economic feasibility.

Another study conducted by HSBC Centre of Sustainable Finance and the Sustainable Digital Finance (2019) examined the impact of blockchain on green finance markets for the third quarter of 2019. The study had focused on opportunities of efficiency and cost savings applicable to green bonds.

The gathered data for the quantitative and comparative study conducted is based on interviews with experts and various literature reviews. It uses data collected from previous research, specifically focusing on reviews conducted over the 2018–2019 period. These literature reviews present different typical fees that are used in this study, in addition to previous cases that focus on comparative analysis between non-DLT and DLT processes. DLT refers to Distributed Ledger Technology, which is a distributed register that uses a network of multiple nodes to maintain and validate transactions in a decentralized way. Blockchain technology is a type of DLTs that ensures validation and storing of transactions in blocks.

Similarly, Haahr et al. [30] highlighted the impact of the blockchain implementation on the green bond's market. Actually, it states that investor confidence, capital flows facility and the cost of reporting are remarkably more affected by blockchain than issuance costs. According to multiple use cases such as Proof of Impact Tokens by UNDP's initiative, Clean Air LifeTokens and TreeCoin by Global Mangrove Trust, investor confidence increases significantly. In fact, these tokens use Internet of Things (IOT) technology, Artificial Intelligence (AI) monitoring or Global Positioning System (GPS) to track real-time green performance, ensuring transparency and enabling donors to see directly the impact of their contribution on the environment.

The authors concluded by asserting that DLT is highly credible and effective in the green market. Even though research on this technology is still in its early stages and there are challenges fitting DLT into traditional banking models, this study urges financial institutions to act and adopt blockchain for their sustainable products. Doing so, it presents an opportunity to increase cost savings, to move to the next stage and open up to new markets with stronger security systems and greater transparency to sustainability.

Bauer et al. [31] outlines the results of a quantitative cost impact analysis of replacing existing corporate bonds with smart bonds. Their study aims to clarify the defined efficiency benefits of distributed ledger technology (DLT) for asset managers, banks and custodians. Therefore, this study proposes a methodology to assess the impact of blockchain-based infrastructure on bond trading costs, particularly focusing on the determinants affecting this cost change.

Due to limited data on tokenized assets, the authors have gathered qualitative data rather than quantitative data. In fact, they based their comparative analysis on interviews with experts such as banks, custodians, etc. in order to identify cost dynamics factors on non-DLT and DLT based assets. Publicly shared quantitative data of traditional exchange market has been used to estimate the initial cost of assets before the intervention of block-chain technology.

The authors examined the DLT-based capital market infrastructure at three time windows: Today (2023, according to the study), 2026 as well as 2028. Because of the modest volume of smart bonds issuance, the DLT-based capital market infrastructure is still in its infancy and therefore has limited economies of scale. Therefore, they evaluated cost in short-term (2026) and long-term (2028), distinctly. This study is conducted on the following underlying assumptions: robust legislative and legal framework, adoption of blockchain technology by the market, accessible technological infrastructure, minimal blockchain fees, and no accounting for expenses on fiat money custody.

Firstly, this study defined different processes in front office, middle and back-office. Right after processes identification and data gathering, researchers calculated the costs of non-DLT processes and estimated the costs of DLT based processes for today, 2026 and 2028. After determining the costs, they compared non-DLT with DLT processes costs. The authors concluded that bonds cost can reach up to 85% down in 2028. In fact, they indicate that blockchain technology had a significant cost impact especially in clearing, settlement and

asset servicing activities. They highlighted that costs after tokenization decrease to the same cost value in both scenarios. However, variability manifested itself when comparing Non-DLT based infrastructure costs, which differs depending on the scenario.

Furthermore, the study deducts certain cost saving shifters and showed its sensitivity to the following determinants listed in Table 1 below:

In the light of these results, the authors encourage market participants to assess and take the appropriate action of implementing DLT infrastructure, at an early stage. They believe that markets should take advantage of different features of DLT to reach gains efficiency.

Conversely, reports on Carbon Credit Tokenization, such as PwC (2024), discussed the environmental benefits of tokenizing carbon credits and the potential advantages for banks in terms of trading efficiency and liquidity.

In this regard, the purpose of our study is to examine how the trading process of the investment bank ODDO BHF would be affected by the adoption of Ethereum blockchain technology. We seek to examine the effects of tokenizing ODDO BHF's financial assets on trading costs in light of the organization's emphasis on cost minimization and sustainability. We also aim to assess the possible effects of this technology on the bank's environmental sustainability reputation.

All the stated studies above have pointed to the positive impact of blockchain on the trading processes of different types of bonds. They succeeded in showing a significant cost-reduction estimation by different methodologies. However, these studies have not considered the environmental impact of tokenizing traditional financial assets. Nonetheless, some findings provide valuable insights, as green bonds or carbon credits tokenization could serve as potential offset mechanisms for emissions arising from high-energy blockchain infrastructures, particularly those using Proof-of-Work consensus mechanisms. Other studies have called for urgent attention to carbon emissions caused by blockchain. In response, we dedicated the second sub-section of this paper to see the assessment of blockchain carbon footprint in different previous studies.

The impact of blockchain on environmental sustainability

In order to comply with international climate targets of reducing carbon emissions, blockchain technology has emerged as a promising tool for carbon management. Its main contribution lies in its ability to facilitate emissions trading platforms, enabling secure, transparent, and tamper-proof transactions. It eliminates intermediaries, thereby reducing transaction costs and increasing efficiency. By enabling a distributed carbon registry, blockchain connects carbon asset management to emissions

Table 1 Cost reduction shifters

Determinants	Sensitivity analysis
Issuance Volume	Economies of scales: The more bonds are issued, the greater cost reduction from conversion to DLT
Maturity	The longer the bond's maturity, the higher total cost reduction because of higher recurring cost within the middle, and back office
Interest payment frequency	Due to the significant decrease in process costs of corporate shares and asset servicing, the greater the interest payment frequency we have, the higher the savings potential are
Investors number	As the number of investors increases, the potential for savings is greater because of cost reductions in clearing and settlement processes, corporate shares, and asset servicing
Origin of investors	Origin of investors can have an impact on cost reduction through DLT. In fact, processes costs when dealing with foreign-based investors, lead to higher cost savings particularly for corporate shares and asset servicing processes
Trading turnover	The higher the trading turnover is, the higher the savings potential are because of a decrease in clearing and settlement costs

Source: [31]

trading systems, ensuring accurate carbon accounting and better control of greenhouse gas emissions [32].

Blockchain technology has the potential to revolutionize carbon markets by automating transactions, reducing costs, and improving transparency. It provides a secure, decentralized, and verifiable framework that not only improves the efficiency of carbon credit trading but also enhances the accountability of businesses and governments in combatting climate change [33, 34].

One of the key areas in which blockchain has a significant impact is the issuance and management of green bonds. According to Ronaghi & Mosakhani [35], it provides a secure and tamper-proof digital ledger that prevents fraudulent transactions and builds trust in green financial instruments, thus promoting a more robust and responsible green financial ecosystem [36].

Blockchain has demonstrated its potential in supply chain management, energy sector, carbon management, and more. It aligns supply chains with the principles of the circular economy, emphasizing the extension of product lifecycles, waste reduction, and resource optimization. In the energy sector, it is used to improve carbon credit trading by ensuring transparency and preventing double counting of emissions. In the energy sector, it facilitates peer-to-peer energy trading, optimizes the issuance of Renewable Energy Certificates (RECs), and improves grid management by enabling secure and decentralized coordination between stakeholders.

Blockchain also enables the tokenization of energy assets. This involves representing energy in the form of tradable digital tokens. These incentive models allow consumers who have reduced their energy consumption or carbon footprint to receive tokens or cash rebates [37]. Not only does this encourage sustainable energy consumption, but also greater participation in decentralized energy markets.

Since the introduction of Blockchain, energy and carbon emissions of Bitcoin and Ethereum blockchains stand among researchers' main areas of interest. In fact, studies on Bitcoin and Ethereum (Before the merge) are conducted to assess the significance of their carbon footprint because they were both following the Proof of work (PoW) consensus mechanism. Other studies have focused on comparing Ethereum's carbon footprint before the merge, during the time when PoW was in use, and after the merge, during the transition to Proof of stake (PoS).

Proof of work (PoW) introduced and popularized by Bitcoin, is the most popular consensus and is considered the most efficient one. Miners need to do computational work to find the nonce that can solve a specific mathematical problem related to the block hash. However, the major drawback of PoW is the computational energy that it needs to find the nonce, which is highly costly and

harms the environment. Many blockchain networks have PoW-based architecture, such as Bitcoin and Ethereum.

On the other hand, with Proof of Stake (PoSPo's), nodes are known as validators that forge blocks instead of mining them. Choosing the validator is done semi-randomly because it considers the size of the stake of each validator as a likelihood determinant.

The merge is an upgrade in the Ethereum mainnet by switching from proof-of-work to proof-of-stake on September 15, 2022. This merge comes to respond to PoW problems of computational costs and has significantly lowered the energy consumption of the network by approximately 99.95%. In view of this, Ethereum is considered now as a green blockchain (Ethereum, 2024).

Recent years have witnessed a remarkable evolution in the discourse surrounding the environmental impact of Bitcoin mining, evidenced by a substantial increase in both the quantity and depth of research on the topic. The enormous energy consumption of Bitcoin mining was shown by early estimates, such as those made by Dwyer & Malone [38], which placed the annual range between 0.88 and 87.60 TWh. A number of other sources, such as Kampl [39], Malmo [40], Deetman [41], and Zohair [42], also made more contributions that enhanced these estimates. However, they suggested a more precise range, placing Bitcoin's annual energy consumption between 1.88 and 7.8 TWh.

Auer [43] argues that proof-of-work Bitcoins are inherently costly and may lead to decreased liquidity as block rewards diminish. The author suggests that maintaining payment finality could become increasingly time-consuming and expensive, concluding that it is inherently expensive and unsustainable in the long run.

In this line of thought, Rauchs et al. [44] offered a thorough analysis of the geographic distribution of miners, providing a further comprehension of Bitcoin's carbon intensity and leading to a weighted carbon intensity of 475 gCO₂/kWh. By calculating the carbon footprint of Bitcoin mining using both IP (Intellectual Property) and mining pool methodologies, Stoll et al. [45]. made an important contribution. These two studies estimated the carbon footprint of Bitcoin mining were 22 MtCO₂ and 22.9 MtCO₂, respectively.

According to Krause & Tolaymat [46], four cryptocurrencies (Bitcoin, Ethereum, Litecoin, and Monero) had a carbon footprint ranging from 3 to 15 MtCO₂ between January 2016 and June 2018. This study highlights, as well, the importance of temporal changes in emissions because of several factors. Actually, the location of a mine can have a significant impact on carbon footprint since distinct energy mixes among countries lead to different emissions. For example, China has four times more CO₂ emissions than Canada because of the increased number of Bitcoin miners there. Hence,

moving mining activities to nations like Canada that have cleaner energy sources lowers emissions. In addition, temporal changes in carbon emission show how dynamic bitcoin mining is and how things like energy prices and progress in technology can affect it.

Examining data from 2017, Mora & al. (2018) indicated that the use of Bitcoin emitted 69 MtCO_{2e}, with significant variation because mining occurs in different countries with different energy profiles and hardware choices. Given that Bitcoin accounts for a comparatively modest percentage of worldwide cashless transactions (0.033% in 2017), its carbon impact is important. The study predicts that Bitcoin's cumulative emissions could exceed standards linked to major global warming (2 °C) in 11 to 22 years if its adoption follows the development trends of other technologies. This shows how important it is to have sustainable mining methods for cryptocurrencies in order to lessen their negative environmental effects.

De Vries [47] reported that the amount of e-waste produced by mining rigs¹ was around 10.95 metric tons or 135 g every transaction, which provides insights into the substantial amount of e-waste generated by these machines. Recently, Sarkodie et al. [48] used econometric and machine learning techniques to study the effects of Bitcoin on a dataset from July 18, 2010 to December 04, 2021, with particular focus on its energy usage and carbon emissions. Their study emphasized the impact of technical factors of Bitcoin, such as hash rate and mining difficulty, on energy consumption and carbon emissions.

Neumüller [49] found that Ethereum mining CO₂ emission is estimated to be 27.5 million tons of carbon dioxide MtCO_{2e}, since the introduction of the network, using PoW, until the transition to the Proof of Stake consensus mechanism. The study shows that 60% of the cumulative emissions occurred during 2021 and 2022. This emission level, which represents 0.06% of greenhouse gas emissions in 2020 is comparable to the annual emissions of some countries such as Honduras (27.67 MtCO_{2e}) and Lebanon (28.87 MtCO_{2e}), in addition to the video gaming industry CO₂ emission, in the United States (24 MtCO_{2e}).

To determine how emission estimates could fluctuate in response to adjustments to important factors, Neumüller [49] carried out a sensitivity analysis. Because of the difficulties in accurately predicting Greenhouse Gas (GHG) emissions, the study evaluated a range of possibilities that included three different scenarios. In the worst-case scenario, all miners were assumed to use coal-fired power only, whereas in the best-case scenario, all miners were expected to use hydropower completely. Between these two extremes lay the most trustworthy assessment. These scenarios demonstrated how there can be a

significant variation in emissions at any given amount of electricity use, depending on the energy source supplying the network. Ethereum's emission debt in the hydro-only scenario is estimated to be around 1.2 MtCO_{2e}. In the coal-only scenario, this number rises by more than 47 times to 58.3 MtCO_{2e}.

These scenarios can be illustrated by examining various countries, where the energy mix used in each affects its carbon emissions. For instance, Kohler and Pizzol (2019) found that regions such as Quebec, Iceland, and Sichuan contribute to low carbon emissions because of their cleaner energy mixes. In contrast, Alberta and Russia, despite having a similar share of mining activity in 2018, exhibit higher carbon emissions.

These studies underscore the significant environmental impact of PoW mining. This points to the urgent need for policy interventions, technological innovations, and a shift towards more sustainable blockchain technologies to mitigate the extensive carbon footprint and energy consumption associated with different blockchains. A marked example of this effort is the Ethereum network's initiative to transition to PoS, a more sustainable consensus mechanism.

Since the Merge, much research has been carried out to investigate how well the Ethereum blockchain handles the sustainability concerns about its former Proof-of-Work consensus process.

While the final pre-Merge GHG emissions of Ethereum network is estimated at 10.3 MtCO_{2e}, on September the 9th, 2022, Neumüller [49] provided an estimate of approximately 2.8 kilotons of CO_{2e} (KtCO_{2e}), based on the most recent post-Merge. In spite of a notable rise in the quantity of Ethereum beacon nodes, these approximations showed a remarkable decrease of almost 99.97%. In addition, according to the most recent estimate of the electricity mix, the network is currently powered by around 48% sustainable energy (32% renewables and 16% nuclear), with wind power accounting for 12% of total renewable energy. Natural gas, coal-fired power, and oil made up 30%, 19%, and 3% of the remaining 52% of fossil fuel production.

Similarly, Hanzo [50] to SG-FORGE presents a carbon footprint methodology of blockchain-based financial instruments. This study is mainly made to propose a framework for the measurement of the carbon footprint based on gas estimation of tokenizing financial instruments, after Ethereum's transition to PoS. Additionally, it seeks to identify factors that significantly increase greenhouse gas emissions in relation to different bond characteristics. To determine the carbon footprint per unit of gas, this study is founded on the approach of Crypto Carbon Rating Institute (CCRI, 2023).

The first step is measuring the power and electricity consumption of a single node. Multiple approaches can

¹ Mining rigs: Specialized computers, dedicated to transaction mining in blockchain.

be used. CCRI and Cambridge University applied the bottom-up approach, which estimates the Upper bound, lower bound and best guess of power and energy consumption according to clients' (the validator nodes) preferences to different hardware. SG-FORGE study uses the Upper bound, lower bound and best guess values of CCRI benchmark as data to continue the rest of the work.

Hanzo [50] aimed to estimate CO₂ emissions arising from the Ethereum's PoS network. To this end, it is important to multiply the electricity consumption of the network, calculated in the previous step, by a carbon intensity factor adjusted to the network nodes' geographic distribution.

To measure carbon intensity first, the authors gathered data for carbon and node distribution, respectively, from Our World Data and Migalabs for the 2022 to 2023 period. To evaluate what can affect the amount of CO₂ emitted in a smart contract, the authors tested four different scenarios of bond in three situations (upper bound, lower bound and best guess).

Depending on the particularity of each bond scenario, the study pointed out that the smart contract's gas usage changes. The study showed that energy use is mostly found in SG-Forge's specialized infrastructure, which includes cloud and node services, for a limited number of tokenizations. On-chain activities, however, use more energy and release more CO₂ in scenarios such as the fourth one, where the number of issuances is much higher. This is especially true in the top bound of the scenario.

The on-chain emissions on Ethereum range from 1.2 kgCO_{2e} to 161.4 kgCO_{2e}. This is comparable to driving 6 km by car (or 19 h of video streaming) and, in the worst scenario, driving 742 km by car. This shows the effective support of the Ethereum blockchain for financial products, especially after its transition from PoW to PoS.

The study, however, has certain drawbacks. Absence of accurate and reliable data is one of its limitations. The authors indicated that even with the significant advancements in the gathering of data on the number of active nodes, their best estimate for node power and energy usage still comes from a theoretical proxy for hardware (a bottom-up method). Another constraint is the ignorance about yearly events that can have a major impact on carbon intensity, like the war in Ukraine and worldwide trends of countries decarbonizing their energy mixes.

Overall, Hanzo [50] believes that blockchain technology maintains considerable potential in lowering climate change. This is because it can improve funding for the transition to an eco-friendly economy and allow for a more transparent approach to tracking and handling of ESG Key Performance Indicators (KPIs) related to financial instruments. Additionally, these researchers think that blockchain-based market infrastructures,

especially using PoS, can have a lower carbon footprint than existing infrastructures.

Empirical design

This study aims to determine if there is an impact of tokenization on bond costs and on environmental sustainability. First, because of the novelty of tokenization and smart assets topics, we observe that there is a remarkable lack in the availability of complete related datasets. To conduct this comparative analysis, our study uses the methodology of the sukuk bond of Khan et al. [29]. We estimate smart bond costs through the code of its smart contract, which involves all the trading process phases from Issuance to Settlement. We apply this approach on one of the offered products of ODDO BHF: the High Yield Bond DP-EUR.

Second, as we are working on issuing the smart bond on Ethereum blockchain, we can assess tokenization's carbon emission in two different phases: Ethereum before and after "The merge". As a starting point, we deduce the carbon emission of our smart contract from the estimated gas consumption in the previous section, by following the methodology of Hanzo [50] to SG-FORGE and CCRI (2023). Right after that, we discuss and compare the carbon emissions of the bond, for both cases, by showing its equivalent in daily activities.

General presentation of ODDO BHF Group

ODDO BHF has been the oldest independent financial group in Europe for more than 170 years. It originated from a German bank that specialized in Mittelstand and a French family firm that was established on the shoulders of five generations of stockbrokers.

Bearing on a strong investment in market expertise, ODDO BHF operates four major businesses: private banking, asset management, corporate and investment banking, and banking services and technologies. With 2,500 employees, 1,300 in Germany, Switzerland, and 1,000 in France and Tunisia and over 142 billion in customer base. The ODDO family owns 60% of the company's capital, while 30% of shareholders are its employees, giving the business a distinctive shareholder structure. The teams' sustained participation is ensured by this cooperation rationale. ODDO BHF reported 781 million euros in net banking income and more than 1,064 million euros in equity, in 2021.

ODDO BHF has more than 60,000 clients, institutional investors, companies, distribution partners and large private clients whose assets the group advises, manages and invests, a total of 142 billion euros.

Today, the ODDO BHF group is present on four continents in 11 countries with approximately 29 locations. Indeed, the group, today, is internationalizing its management activities, which now represent around 4.5 billion

euros in assets. The capital raised is mainly invested outside France, mainly in Europe, but also in other parts of the world.

With the goal of meeting the needs of its clients, ODDO BHF is engaged in the development of innovative offers and products. The institution has introduced funds like the ODDO BHF Artificial Intelligence fund, which uses AI in its investment process, and Alpha Intelligence Capital (AIC), a venture capital fund devoted to artificial intelligence run by acknowledged experts. Additionally, ODDO BHF often makes investments in FinTech and InsurTech, mostly in France and Germany. Companies like Lydia, Simplesurance, and Wizbii are among those with significant benefits and ambitious business models that the fund targets. By providing training on blockchain technology and artificial intelligence, as well as putting out calls for creative projects, ODDO BHF is fostering a digital culture that will revolutionize the working life of its staff employees. Because of one of these initiatives, The Ladies Bank was established as a platform designed to help women build their networks and manage their assets more skillfully. Furthermore, ODDO BHF has forged multiple academic partnerships, in order to reinforce its technological and educational projects.

ODDO BHF is implementing a green and eco-friendly strategy aimed at environmental sustainability. It ensures that its employees are highly motivated to make an internal commitment to the environment by changing their behaviors, with the goal of reducing both their own environmental impact and that of the Group. Today, ODDO BHF offers a wide range of products that meet its green strategy.

In 2009, Philippe ODDO took the initiative to enlarge the Group's presence in Tunis. At that time, the managing partner founded the ODDO Research Institute, in order to help the ODDO BHF Group's growth. This strategic decision-making plan aligns with the Group's substantial research expenditure. ODDO BHF Tunis is a forward-thinking firm that is making waves in the Tunisian market. It plays a significant role in the expansion of the Group.

The OBAS (ODDO BHF ASSET SERVICING) team provides a range of front-to-back solutions, such as IT, administrative, and operational outsourcing for private banks and insurance firms, as well as deposit account administration for asset managers. Furthermore, it serves as a designated custodian for both listed and unregistered funds and maintains the shareholder registries for these funds.

Data collection

The choice of a bond as the tokenized financial asset to study stems from a lack of previous studies on the tokenization of other financial assets, such as shares, and from a lack of advanced technical expertise required to code a more complex smart contract for other asset types. According to an expert in smart contracts, the smart contract used in this study is suitable for academic and estimation purposes.

To avoid additional complexities related to the underlying asset and to make the case study more broadly applicable to other financial institutions, we selected the ODDO BHF Euro High Yield Bond, as it is a product with characteristics that are comparable to similar high-yield bonds offered by other institutions.

To test the impact of tokenization on bond costs, the primary sources of our data are ODDO BHF documents and Ethereum open sources like Etherscan. We gathered data on the bond under analysis from SICAV ODDO BHF 2023 annual report. It covers various High Yield Bond fees such as transaction, management and auditing fees. Table 2 below presents the bond's primary characteristics:

From the World Bank guide on international bond issuance (2019), we also extracted different cost rates, like the rating fee rate. In order to determine the gas costs associated with the smart contract, Etherscan is used to obtain the average gas price calculated in Gwei. This comparative study is referenced to October 31, 2023, since availability of research and data on smart bonds is limited to this date.

- Our study will be based on the following underlying assumptions:
- Costs are computed for one investor per smart contract.
- We will consider that each investor has an investment of 10 M (the minimum investment).
- Article 2(4)(a) of Markets in crypto-assets regulation (MiCa, 2023) states that this type of token, which is classified as a financial instrument, does not fall under MiCA regulation. Instead, it falls under the same regulation as the underlying product, which is the European Directive 2014/65/EU called the second Markets in Financial Instruments Directive

Table 2 ODDO BHF high yield bond DP-EUR Key Features

Fund Name	ODDO BHF high yield bond DP-EUR
ISIN Code	LU0456627214
Minimum investment	10,000,000 EUR
Net asset value per unit (31/10/2023)	10,59 EUR
Subscription fees	100 EUR
Management fees	0,45% of Net asset value
Administrational costs	0,45% of Net asset value
Transaction fees	0,13% of Net asset value

Source: (ODDO BHF, 2023)

Table 3 ERC-20 functions

Function	Description
TotalSupply()	It shows how many tokens exist in total for this specific type of tokens
BalanceOf(address account)	It indicates how many tokens the account holds
Transfer(address recipient, uint256 amount)	It is useful to send tokens to someone else
Allowance(address owner, address spender)	Refers to the number of tokens a spender is allowed to spend on behalf of someone else
Approve(address spender, uint256 amount)	This function allows an account owner to give permission to or approve another account, which is the spender to spend a certain amount of their tokens
TransferFrom(address sender, address recipient, uint256 amount)	This function allows a spender, who has been approved by the owner, to transfer tokens from the owner's account to someone else's account

II (MIFID II), as transferable securities in Article 4(1)(44) of MiFID II (2014). The EU is still in the experimental phase of incorporating tokenized bonds and shares under MIFID II. Regulators are working on updating MIFID II directives through various actions, such as the DLT Pilot Regime, which is a sandbox that allows experimentation with financial services under MIFID II on new technological infrastructure. **In light of this, the study will be based on the assumption that the token falls under solid regulation.**

To determine the carbon emission of the bond, a comprehensive data extraction process was undertaken. It involves the collection of information from various sources, including Etherscan, the CCRI report, Cambridge Center for Alternative Finance (CCAF), Migalab, and global world data. The Cambridge blockchain

Table 4 BondToken functions

Function	Description
Addissuer	It enables the contract owner to add the issuer address. This function will give access to the issuer to do certain actions in the contract
Addregulator	Similar to the addissuer, this function is used to add a regulator in order to do regulatory measures
Getprice	This function will return the price of the token
Addinvestor	It allows for adding one or more investor addresses, who are interested in participating in smart bond exchange markets. This will allow investors to access the trading platform
Onboardclient	This function provides an automatic update to client data structure as well as facilitates customer relationship management
VerifyKYC	Regulators verify and validate customer's KYC using this function

Source: Author

Table 5 Bondtrading functions

Function	Description
Placeorder	Allows issuers and investors to place sell and buy orders for bond tokens. Users specify the order type (buy or sell), the amount of tokens they want to trade, and the price per token. If price and amount > 0, the order will be created and added to orderBook
Executetrade	This function matches a buy and a sell order from the orderBook

Network Sustainability Index (CBNSI), launched by CCAF, presented updated estimates of power and energy consumption prior and posterior to the merge, as well as historical data for the number of active nodes in Ethereum.

Etherscan's contribution came in the form of daily Ethereum gas consumption data, represented in a bar chart format. Migalab's data was instrumental in understanding the geographic distribution of nodes. Additionally, global world data was leveraged to ascertain electricity carbon intensity. We conducted the comparative analysis on two specific dates: October 31, 2023, and May 15, 2022. In fact, the chosen dates were strategically set to create a significant time gap from the Ethereum transition date of September 15, 2022, in order to minimize the potential impact of news of the merge on the parameters under examination.

Methodology

Estimation of the smart bond costs

To write the smart contract code by solidity language, we have used Remix programming tool. Remix IDE is one of the most useful tools for writing Ethereum smart contract codes. It makes it easy for developers to create, test, and modify their contracts. Remix allows them to write Ethereum contracts in Solidity and test their functionality, as well as estimate its gas requirements. Because of its simple form and extra features, it is ideal for beginners as well as experts. Additionally, Ethereum developers worldwide will find it quite easy as it can be used both online and on a computer.

In this study, we aim to estimate the gas consumption of transforming the bond to an ERC-20 token, issuing it, and coding all the trading process steps.

The code includes five parts: The ERC-20 token interface and four smart contracts for issuance and pre-trade, trade execution, trade clearing and settlement. As it is a comparative study, we present a simplified code that contains all main functions required for cost estimation. This code is mostly useful in academic and research contexts. Although it provides an elementary sample for a smart bond, further development, and improvement are important to implement a smart contract in a live blockchain.

ERC-20 Interface code

Of all ERC standards, we have chosen the ERC-20 for our bond, as it is the most suitable for our study which proposes a simple research prototype for bond tokenization. The following code 1 illustrates the set of standards or functions related to ERC20. It is considered as a template that must be followed to make our tokenized bond. This code does *not* consume any gas.

Code 1: ERC-20 interface

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

interface IERC20 {
    function totalSupply() external view returns (uint256);    ⛜ - gas
    function balanceOf(address account) external view returns (uint256);    ⛜ - gas
    function transfer(address recipient, uint256 amount) external returns (bool);    ⛜ - gas
    function allowance(address owner, address spender) external view returns (uint256);    ⛜ - gas
    function approve(address spender, uint256 amount) external returns (bool);    ⛜ - gas
    function transferFrom(address sender, address recipient, uint256 amount) external returns (bool);

    event Transfer(address indexed from, address indexed to, uint256 value);
    event Approval(address indexed owner, address indexed spender, uint256 value);
}
```

Table 3 below presents the ERC-20 standard functions and their utility:

BondToken code

The following smart contract “bondToken” serves to create the token, issue it on the exchange market and execute the pre-trading phase. Based on the ERC-20 implemented infrastructure, the contract provides specific

functionalities to ensure compatibility and interoperability with existing token-based systems.

The BondToken SC introduces functions for client management, such as onboarding, KYC verification, and risk assessment, in order to ensure regulatory compliance and mitigate risks. Furthermore, the contract gives the issuer access to conduct operations of client whitelisting for token purchases. In addition to basic token management, the

contract facilitates role-based access control, by allowing distinct permissions for regulators, issuers, and investors.

BondToken functions are summarized in Table 4 below.

The following code 2 illustrates part of the SC “BondToken” code by highlighting its main functions:

Code 2: Bondtoken smart contract code

```
contract BondToken is IERC20 {
    //the rest of the code ...
    // Owner can add issuers
    function addIssuer(address _issuer) external onlyOwner
    // Owner can add regulators
    function addRegulator(address _regulator) external onlyOwner

    // Function to get the price of the bond tokens
    function getPrice() external pure returns (uint256)

    // Owner can add investors
    function addInvestor(address _investor) external onlyOwner

    // Client onboarding function
    function onboardClient(address _client) external onlyIssuer

    // KYC verification and risk assessment function
    function verifyKYC(address _client, uint256 _riskScore) external onlyRegulator

    // Whitelist a client for bond token purchases
    function whitelistClient(address _client) external onlyIssuer

    // Remove a client from the whitelist
    function removeClientFromWhitelist(address _client) external onlyIssuer

    // Check if a client is whitelisted
    function isClientWhitelisted(address _client) external view returns (bool)
}
```

Table 6 ClearingContract functions

Function	Description
Cleartrades	Calls Investor and issuer's data and verify if they meet the matchCriteria
TradesettlementStatus	If trade is cleared, this function returns the trade ready to settlement

Bondtrading code

The BondTrading smart contract represents the trade execution phase of the process. It defines functions (see Table 5 below) that will enable placing buy and sell orders for issuers and investors, as well as ensuring their matching.

Code 3 includes two main functions, which are PlaceOrder and ExecuteTrade:

Code 3: Bondtrading smart contract

```
contract BondTrading {
  //The rest of the code ...
  // Place a buy or sell order
  function placeOrder(OrderType _orderType, uint256 _amount, uint256 _price) external {
    require(_amount > 0, "Amount must be greater than zero");
    require(_price > 0, "Price must be greater than zero");

    Order memory order = Order(msg.sender, _orderType, _amount, _price, false);
    orderBook[msg.sender].push(order);

    emit OrderPlaced(msg.sender, _orderType, _amount, _price);
  }

  // Execute trades by matching buy and sell orders
  function executeTrades() external {
    Order[] storage buyOrders = orderBook[msg.sender];
    Order[] storage sellOrders = orderBook[msg.sender];

    for (uint256 i = 0; i < buyOrders.length; i++) {
      for (uint256 j = 0; j < sellOrders.length; j++) {
        if (!buyOrders[i].executed && !sellOrders[j].executed && buyOrders[i].price >= sellOrders[j].price) {
          uint256 amount = (buyOrders[i].amount < sellOrders[j].amount) ? buyOrders[i].amount : sellOrders[j].amount;

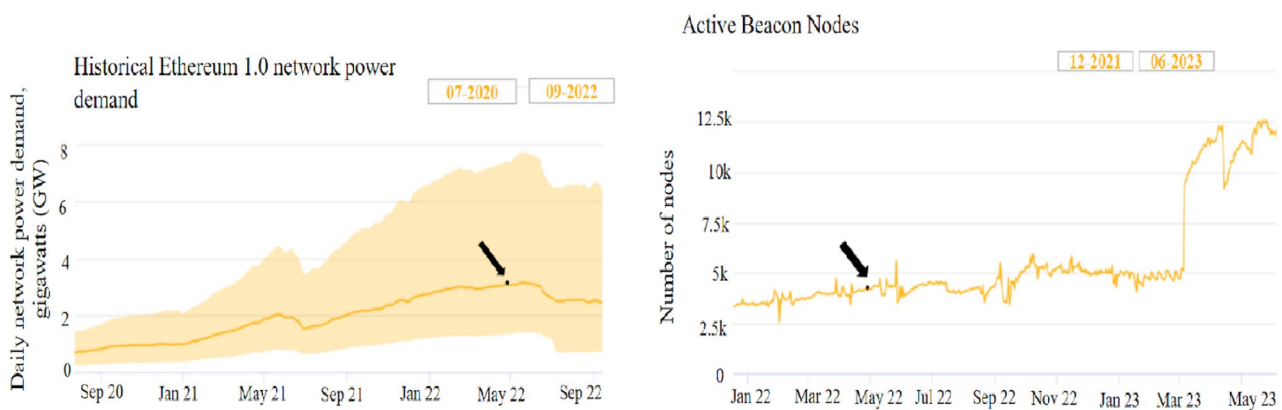
          // Execute trade
          bondToken.transfer(sellOrders[j].trader, amount);
          sellOrders[j].amount -= amount;
          buyOrders[i].amount -= amount;
          emit TradeExecuted(buyOrders[i].trader, sellOrders[j].trader, amount, sellOrders[j].price);

          // Mark orders as executed if fully filled
          if (sellOrders[j].amount == 0) {
            sellOrders[j].executed = true;
          }
          if (buyOrders[i].amount == 0) {
            buyOrders[i].executed = true;
          }
        }
      }
    }
  }
}
```

Clearingcontract code:

This smart contract aims to ensure that trading parties meet the matching criteria. In fact, it interacts with the BondToken and BondTrade smart contracts to get data about both seller and buyer (price, quantity, Unique Transaction Identifier (UTI), etc.) and check if they match, in order to change the trade status to “ready to settlement”. To execute the clearing phase, the following code 4 includes two main functions, which are ClearTrades and TradeSettlementStatus, and one internal function to ClearTrade, which is MatchCriteria.

The following Table 6 defines respectively ClearTrades and MatchCriteria functions:

**Fig. 1** Best guess daily power demand (GW) & active nodes for Ethereum (Before Merge) Source: (CCAF)

Code 4: Clearingcontract smart contract

```

contract ClearingContract {
    BondToken public bondToken;
    //the rest of the code ...
    event ClearedTrade(address indexed buyer, address indexed seller, uint256 amount, uint256 price);

    // Function to clear trades by updating order statuses
    function clearTrades(address[] calldata _buyers, address[] calldata _sellers, uint256[] calldata _amounts, uint256[] calldata _prices) external {
    }

    // Internal function to check if trades match the clearing criteria
    function _matchCriteria(address _buyer, address _seller, uint256 _amount, uint256 _price) internal view returns (bool) {
    }

    function tradeSettlementStatus(address, address, uint256, uint256) external pure returns (bool) {
    }

    // This function is just a placeholder implementation
    return true; // You can customize the logic as per your requirements
    }
}

```

Bondsettlement code:

The BondSettlement smart contract defines the last phase of the trading process, which is trade settlement. We have set the settleTrade function to execute this step and it is illustrated by the following code 5. It interacts with ClearingContract SC, in order to verify that the trade is ready to settlement. If so, the function will transfer bond tokens to the investor and funds to the issuer.

Carbon foot print assessment

To compute carbon emission per gas unit, the first thing to do is determine the daily power demand and energy consumption, per node, for 31/10/2023 and 15/05/2022.

Power demand

Power demand in blockchain networks is defined as the instantaneous total amount of computational power for all miners. It encompasses the energy needed to support

Code 5: Settlementcontract smart contract

```

contract BondSettlement {
    ClearingContract public bondClearing;
    BondToken public bondToken;

    event TradeSettled(address indexed buyer, address indexed seller, uint256 amount, uint256 price);

    constructor(address _bondClearingAddress, address _bondTokenAddress) {
        bondClearing = ClearingContract(_bondClearingAddress);
        bondToken = BondToken(_bondTokenAddress);
    }

    // Settlement function to exchange bond tokens for cash
    function settleTrade(address _buyer, address _seller, uint256 _amount, uint256 _price) external {
        require(_buyer != address(0) && _seller != address(0), "Invalid addresses");
        require(_amount > 0 && _price > 0, "Invalid trade details");

        // Ensure trade has been cleared
        require(bondClearing.tradeSettlementStatus(_buyer, _seller, _amount, _price), "Trade not cleared yet");

        // Transfer bond tokens from seller to buyer
        bondToken.transfer(_buyer, _amount); // Corrected the transfer function arguments

        emit TradeSettled(_buyer, _seller, _amount, _price);
    }
}

```

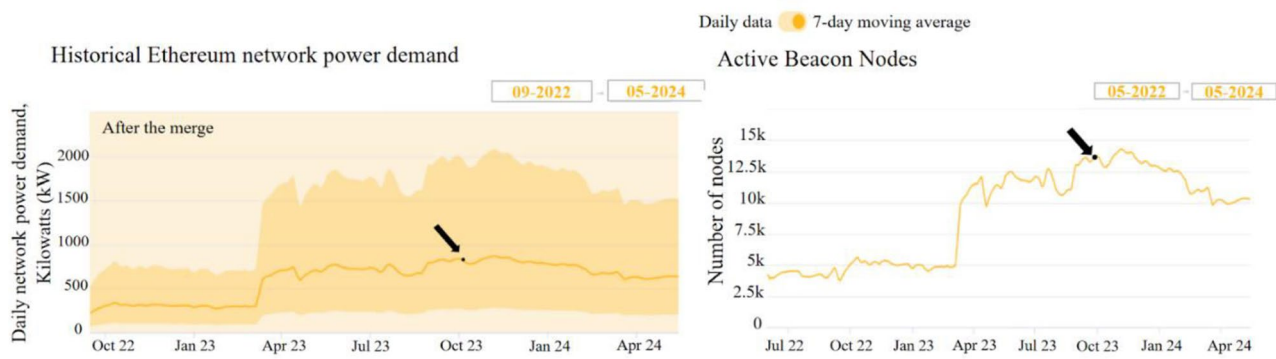


Fig. 2 Best guess daily power demand (kW) & active nodes for Ethereum (After Merge) Source: (CCAF)

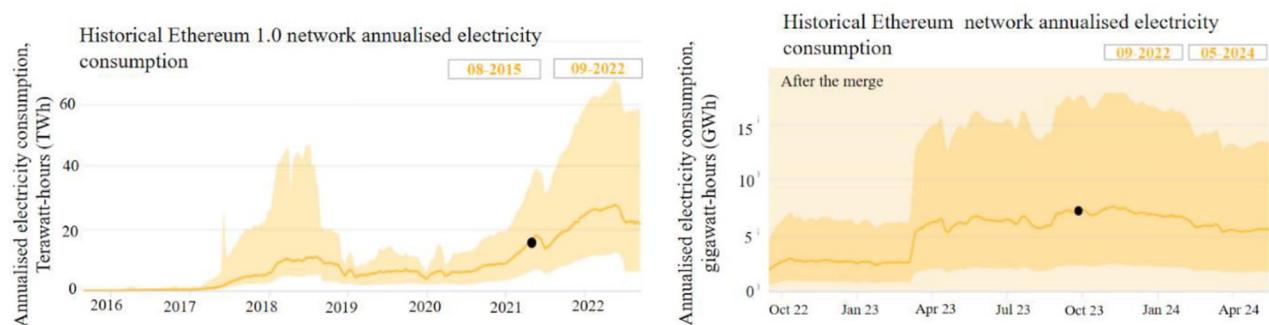


Fig. 3 Estimates of annualized energy consumption before merge (TWh) and after merge (GWh) Source: (CCAF)

Carbon intensity of electricity, 2022

Carbon intensity is measured in grams of carbon dioxide-equivalents emitted per kilowatt-hour of electricity.

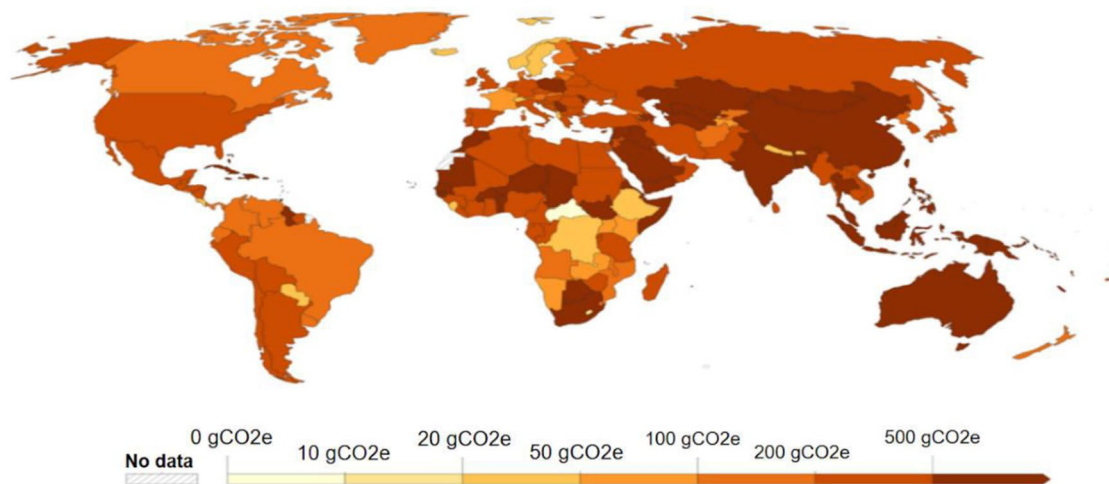


Fig. 4 Carbon intensity of electricity in 2022 Source: (Our World Data)

transactions, smart contracts, and associated processes. Unlike gas consumption, which is specific to a transaction and unaffected by the number of miners, power demand is influenced by the collective computational capabilities of all participants in the network.

To calculate power demand, several studies are built upon CCRI and Cambridge University “Bottom-up” methodology. In this approach, power demand estimation is based on building a basket of real-world hardware with the underlying assumption that miners are rational economic agents that only use profitable hardware. We are going to

Table 7 Carbon intensity of 2022 for node locations

	Country	Nodes proportion per country (1)	Carbon intensity of electricity—gCO ₂ /kWh (2)	Carbon proportion (1)×(2)
1	United states	35.05%	368.10	129.01
2	Germany	17.50%	385.39	67.45
3	France	6.15%	84.88	5.22
4	united kingdom	5.69%	261.15	14.86
5	Finland	4.76%	131.71	6.26
6	Netherlands	4.04%	354.31	14.32
7	India	2.68%	633.40	16.98
8	Canada	2.64%	125.84	3.32
9	Ireland	2.44%	346.43	8.45
...	...	...	...	...
58	Cyprus	0.02%	589.35	0.13
59	Ecuador	0.02%	183.63	0.04
60	Moldova	0.02%	666.67	0.13
			CI_{ETH} = Σ Carbon proportion	315.51

take the power best guess value, on a daily basis, proposed by Cambridge University on 15/05/2022 and 31/10/2023.

- Before merge:

As Ethereum blockchain used to use Proof of work before merge, power consumption was significantly high. The following two-line graphs (Fig. 1) illustrate, respectively, the daily power demand of Ethereum network in gigawatts and the number of active nodes before merge:

According to Fig. 1:

- Daily power demand (15/05/2022) = P = 3.148 GW²
- Active nodes number (15/05/2022) = 3928

Table 8 Gas consumption for the periods 05/21—05/22 and 11/22—11/23

	Before merge (May 21-May 22)	After merge (Nov 22-Nov 23)
Gas consumption	3.55869E+13	3.93731E+13

$$\begin{aligned} & \text{– Daily power demand per node} \\ & (W) = \frac{\text{Daily power demand}}{\text{Active nodes number}} \times 10^9 = \frac{3.148}{3928} \times 10^9 \\ & = 801,425.66 \text{ W.} \end{aligned}$$

- After merge:

After Ethereum transitioned to proof-of-stake, power demand remarkably declined from 2.44 GW on 14/09/2022 (the last day before merge) to around 300 KW. It had, then, a slight rise on the 6th of March 2023, because of upgrading the monitoring network tool and discovering an increase of active nodes number from approximately 5000 to 12,000, as shown in Fig. 2.

According to Fig. 2 above:

- Daily power demand (31/10/2023) = P = 855.9104 kW
- Active nodes number (31/10/2023) = 14,231
- Daily power demand per node (W) = $\frac{855.9104}{14231} \times 10^3 = 60.144 \text{ W}$

Energy consumption:

Energy consumption relates to power demand. It refers to the total amount of energy used over a specific period. In fact, power is considered as the time rate of energy consumption *i.e.* how much energy is consumed at a given point in time. Energy consumption is calculated in the following way:

**Fig. 5** Ethereum daily gas used Source: (Etherscan)

² 1 GigaWatts (GW) = 10⁶ KiloWatts (KW) = 10⁹ Watts.s

$$\begin{aligned} \text{Energy consumption (Wh)} \\ = \text{power (W)} \times \text{Time(h)} \end{aligned}$$

*Wh: Watt-hours

In order to calculate energy consumption in KWh, over one-year period, we adjust the formula as follows:

$$\begin{aligned} \text{Energy consumption (KWh/year)} \\ = \text{power (W)} / 1000 \times 24 \times 365.25 \end{aligned}$$

Accordingly, we determine the average energy consumption:

- *Before merge*: Annual Energy consumption per node
 $= \frac{801425.66}{1000} \times 24 \times 365.25 = 7,025,297.336 \text{ KWh/year.}$
- E_{ETH} = Ethereum network energy consumption = Energy consumption per node $\times$ Ethereum nodes number
 $= 7,025,297.336 \times 3928 = 27,595.37 \text{ GWh} = 27.59537 \text{ TWh}$
- *After merge*: Annual Energy consumption per node
 $= \frac{60.144}{1000} \times 24 \times 365.25 = 527.222 \text{ KWh}$
- E_{ETH} = Ethereum network energy consumption
 $= 527.222 \times 14,231 = 7.5 \text{ GWh}$

We can check our estimations by observing that energy consumption line charts, provided by Cambridge University in Fig. 3, present approximately similar values as calculated above.

Calculation of ethereum annual carbon emission

To measure the annual carbon emission of the network, we relied on the presented model of SG-FORGE that calculates CO_2 as follows:

$$\text{CO}_{2ETH} (t\text{CO}_{2e}) = CI_{ETH} * E_{ETH}$$

* CI_{ETH} : Carbon intensity refers to the distribution of carbon production among the energy mix of a country, region, or network. The energy mix encompasses all sources of carbon such as natural gas, oil, wind, solar, hydro, nuclear, etc.

In our case, we determine CI_{ETH} according to the distribution of Ethereum nodes in every location in the world, and the annual energy consumption of each country in 2022 as shown in Fig. 4 below.

To do so, we used power query as a tool to transform datasets of carbon intensity and nodes distribution for each country in 2022, which are respectively extracted

Table 9 The HYB smart contract gas consumption

Function	Estimated gas used
Bondtoken contract	2,122,409
Bondtrading contract	982,468
Clearingcontract	831,833
Bondsettlement	622,215

Source: Remix IDE

Table 10 Smart contracts gas fees

Function	Estimated gas used	Gas cost (Ether)	Cost in €
Bondtoken contract	2,122,409	0.062	104.637
Bondtrading contract	982,468	0.028	48.437
Clearingcontract	831,833	0.024	41.01
Bondsettlement	622,215	0.018	30.676
Total	4,558,925	0.132	224.76

from Our World data, CCAF and MigaLabs websites, in order to calculate the weighted average.

Table 7 provides the results on carbon intensity of 2022 for node locations, as well as carbon proportions.

Accordingly, we estimate carbon intensity for 2023, which showed a slight increase compared to 2022 and reached 343.222 gCO₂/kWh, despite a decrease in Germany's carbon intensity proportion following the Nuclear phaseout.

As a result, we use the estimated CI_{ETH} value to calculate the annualized carbon footprint of Ethereum CO_{2ETH} :

- *Before merge*:
 $\text{CO}_{2ETH} = 315.51 \times 27,595.37 = 8,706,615.19 \text{ tCO}_{2e}$
- *After merge*: $\text{CO}_{2ETH} = 343.222 \times 7.5 = 2574.165 \text{ tCO}_{2e}$

Annualized gas consumption

To determine carbon emission per unit of gas, we consider the gas consumption of the Ethereum network over a yearly period. In our case, we extracted a daily gas consumption dataset from Etherscan (see Fig. 5 below) and we calculated annual consumption by summing the daily data for two periods:

- *Before merge*: from 01/05/2021 to 30/04/2022
- *After merge*: from 01/11/2022 to 31/10/2023

We noticed that gas consumption did not decrease after the merge. Contrarily, it has slightly increased and remained steady. Ethereum merge focused on reducing energy consumption and carbon footprint, not on reducing gas consumption and speeding transactions. In fact, Gas consumption depends on transactions and their complexity.

We found the following results on gas consumption for the two specified periods in Table 8:

Table 11 Traditional and smart bond issuance expenses

Type	Fees	Fee rate	Fee amount* (€)	
			Before tokenization	After tokenization
Tokenization dependent costs	Initial subscription fees		100	All tokenized and calculated in
	Client administration costs	0.45%	45,000	estimated by gas consumption
	Fund administration costs			of the smart contract
	Regulatory costs			(Table Smart contracts gas fees)
	Transaction costs	0.13%	13,000	
	Total of dependent costs		58 100.00	224.76
Sum of independent costs: 81,356,318,277				
Tokenization independent costs	Rating fees		8872.713407	8872.713407
	Legal advisory		12,983.60487	12,983.60487
	Auditing fees	0.10%	10,000	10,000
	Depository fees	0.045%	4500	4500
	Annual Management Charges	0.45%	45,000	45,000

***Amount of fees:** different expenses are calculated based on the net asset value (NAV) of the bond. NAV refers to the intrinsic value of a share in a fund's portfolio. For one investor, the amount of fees is calculated in the following way:

Table 12 Cost-analysis comparison between traditional and smart bond

	Traditional bond	Smart bond
Total dependent costs	58 100.00	224.76
Total independent costs	81,356,32	81,356,32
Total	139,456,32	81,581,08
Cost reduction rate		41.5%*

* Cost reduction rate = $\frac{(139456,32 - 81581,08)}{139456,32} = 41.5\%$

Carbon emission per unit of gas:

Similar to SG-FORGE, we calculate carbon emission per unit in the following way:

$$CO_{2ETH/GAS} (gCO_{2e}) = CO_{2ETH} / G_{ETH}$$

These are the estimated values of average carbon emission per unit of gas for the two specified earlier periods:

$$- \text{Before merge: } CO_{2ETH/GAS} = \frac{8706615.19}{3.55869E+13} = 0.245 \text{ gCO}_2$$

Table 13 Carbon footprint of HYB smart contract

Function	Estimated gas used	Before merge	After merge
		Carbon footprint (gCO _{2e})	
Bondtoken contract	2,122,409	519,263.53	127.557
Bondtrading contract	982,468	240,368.28	59.046
Clearingcontract	831,833	203,514.28	49.993
Bondsettlement	622,215	152,229.64	37.395
Total	4,558,925	1,115,375.73	298.0239
	Carbon footprint (KgCO_{2e}):	1115.376	0.298
	$CO_2 = \frac{2ETH/GAS \times \Sigma Gas_{(SC)}}{1000}$		

Table 14 Comparison between Carbon footprint before and after merge

	Ethereum 1.0	Ethereum 2.0	Reduction rate
Carbon footprint (KgCO _{2e})	1115.376	0.298	99.975%

$$- \text{After merge: } CO_{2ETH/GAS} = \frac{2574.165}{3.93731E+13} = 6.53788E-05 \text{ gCO}_2$$

Results and discussion

Impact of tokenization on bonds cost

As mentioned before, our main purpose from writing the smart contract code is to estimate execution costs of creating, issuing and trading a smart bond, in order to compare it to how much it costs without tokenization.

Cost estimation smart contract:

As a starting point, we determined costs by deploying each smart contract separately on Remix. Table 9 below includes the gas estimation of our code:

Normally, gas price differs from one transaction to another depending on the smart contract complexity and its execution speed (the higher the gas price is, the more the transaction is prioritized). In our case, we will base our analysis on the average gas price to avoid overestimating or underestimating costs based on short-term extremes. It is calculated in Gwei³ as of October 31, 2023, according to Etherscan:

$$\begin{aligned} \text{Average gas price (31/10/2023)} &= 29.03 \text{ Gwei} \\ &= 29.03 \cdot 10^{-9} \text{ Ether} \end{aligned}$$

³ Gwei: is a sub-unit for Ether, the cryptocurrency of Ethereum: **1 Gwei** = 10⁻⁹ **Ether**.

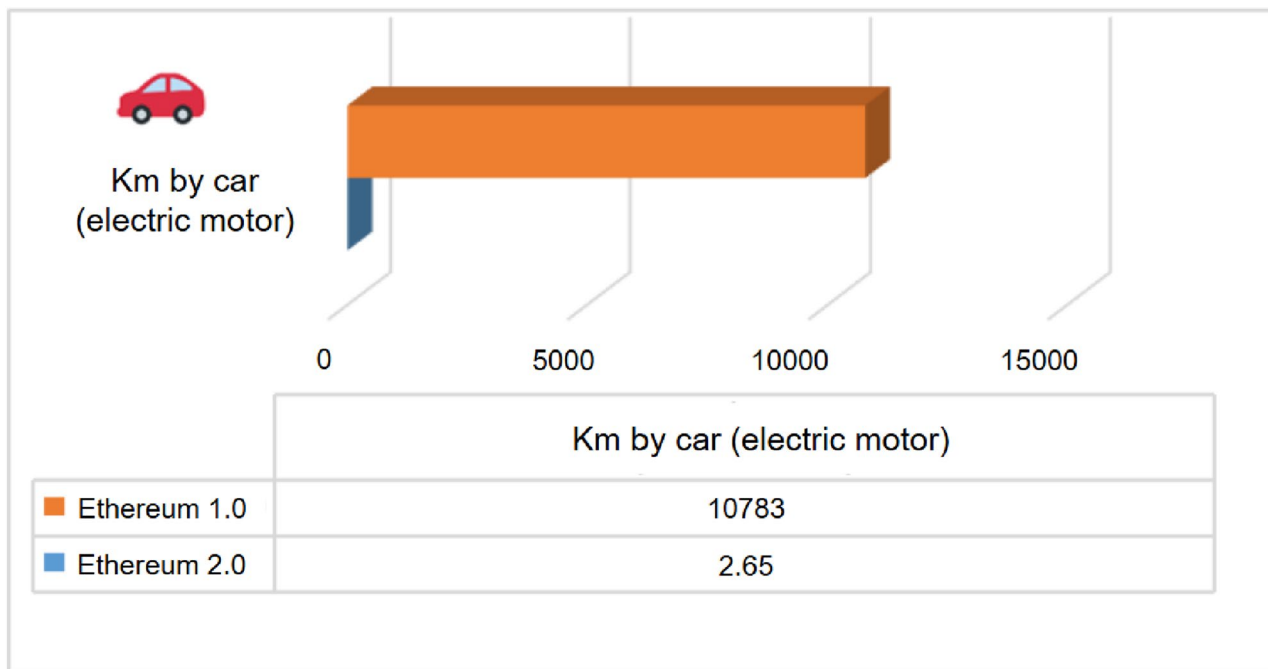


Fig. 6 SC's carbon footprint in equivalent car kilometers Source: (ImpactCO2)

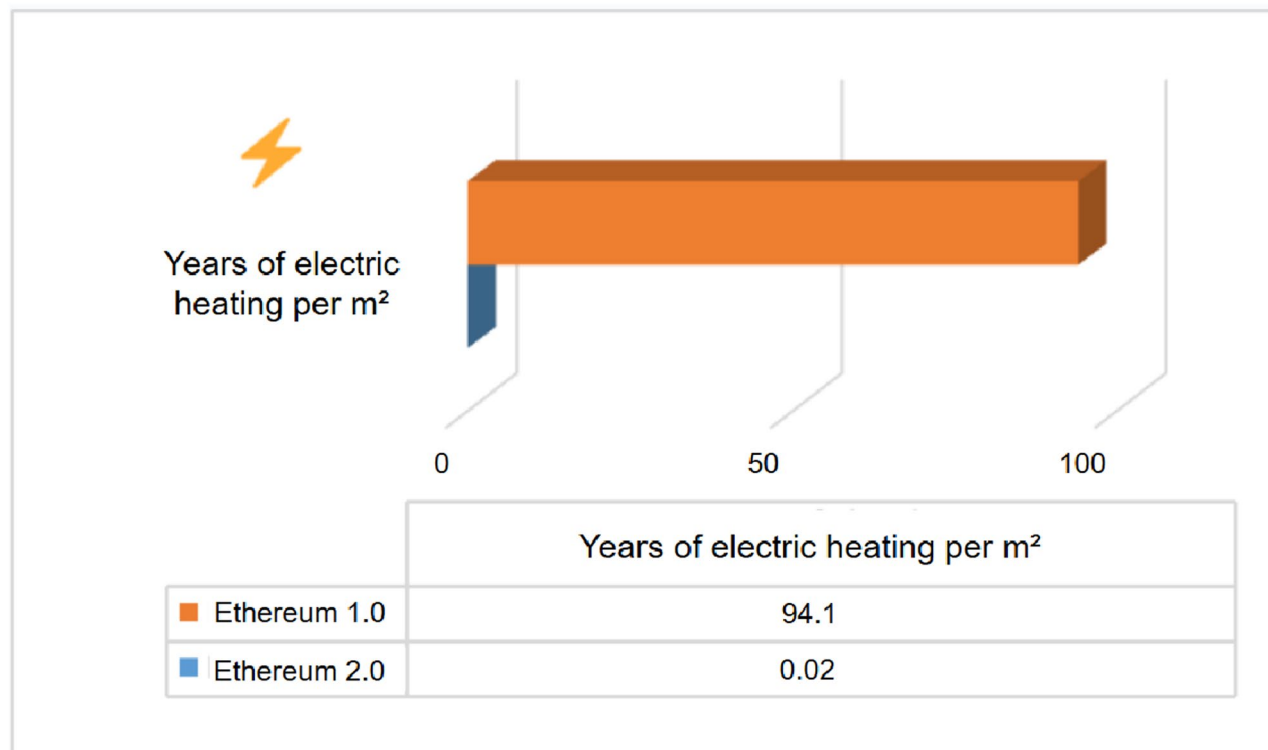


Fig. 7 SC's carbon footprint in equivalent years of electric heating Source: (ImpactCO2)

In order to compare the token's costs to the traditional bond costs, we need to convert all costs to a fiat currency. For our case, we will be using Euros. On 31/10/2023,

the exchange rate was 1700.04 Ether to Euros. Different smart contracts costs are presented in the following Table 10:

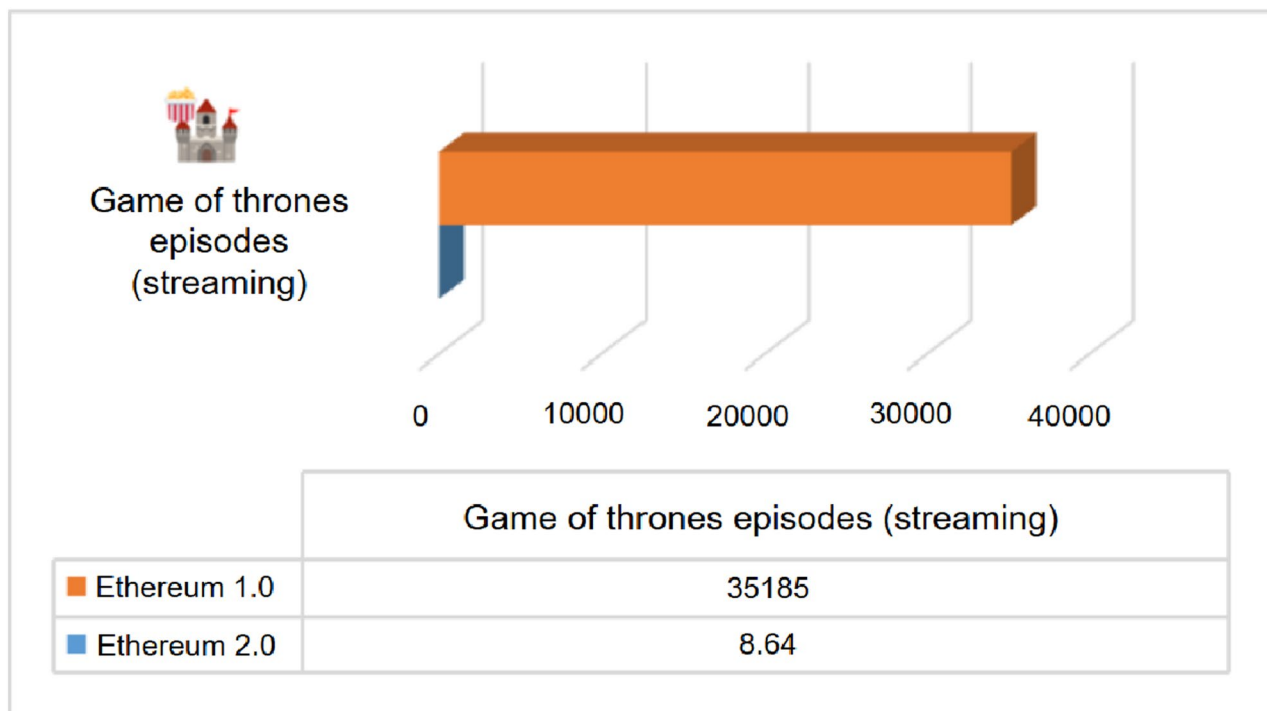


Fig. 8 SC's Carbon Footprint in equivalent Game of Thrones Episodes Source: (ImpactCO2)

Below is an explanatory example of gas costs calculation for the BondToken smart contract:

- BondToken contract gas cost = $2122409 \cdot 10^{-9} \times 29.03 = 0.062$ Ether = $0.062 \times 1700.04 \text{ €} = 104.637 \text{ €}$

Comparative cost analysis

We compare the costs of the “ODDO BHF High Yield DP-EUR” bond, before and after tokenization to assess its impact on the trading process. To begin, we extracted bond-related expenses from different documents, such as SICAV ODDO BHF annual report of 2023, monthly fact-sheets, etc. According to the cost data that we found, and to serve the purpose of this study, we categorized costs into two classes:

- Tokenization-dependent costs: refer to costs associated with activities that will be encoded in the smart contract. In this class, we included three types of fees:
 - Initial subscription fees.
 - Administration fees: This covers three types of costs: client administration costs, funds administration costs and regulatory costs.

- Transaction fees: Cover costs related to the issuance, trade execution, clearing, and settlement.

- Tokenization-independent costs: the rest of the costs are considered unaffected by tokenization and remain unchanged in both situations. In our study, we list rating fees, legal expenses, auditing, and custodial and management fees. In the following Table 11, we present all costs to be considered in this study:

$$\begin{aligned} \text{Fees} &= \text{Fee rate} \times \text{Net asset value (per bond)} \\ &\quad \times \text{Number of invested bonds} \\ &= \text{Fee rate} \times \text{investor's investment} \end{aligned}$$

On 31/10/2023, we have:

- Net asset value (31/10/2023) = 10.59 EUR
- Investment per investor = 10,000,000 EUR
- Number of shares for each investor = $10,000,000 / 10.59 = 944,287.063 \approx 944,287$ shares

As an example, we calculate management fees:

- Management fees = $0.45\% \times 10,000,000 = 45,000$

Table 12 below shows the cost-analysis comparison between traditional and smart bonds and provides the cost reduction rate.

Adoption of the tokenization procedure for the ODDO BHF HYB has significantly contributed to a 41.5% cost reduction, demonstrating the blockchain's powerful impact on cost reduction. This estimate, however, ignores how tokenization may affect the environment, particularly in light of the extensive discussion about the high carbon emissions associated with blockchain technology.

Results from carbon emission assessment

After calculating the carbon footprint of one unit of gas, we compare the amount of carbon emitted by the smart contract of ODDO BHF High Yield Bond before and after the merge and compare these two periods.

As we mentioned previously, Ethereum transition does not affect the gas usage of a transaction. Then, we estimate carbon footprint for both phases based on the function of gas usage estimated in the cost-benefit section. The results are included in Tables 13 and 14, and supported by Figs. 6, 7 and 8 as follows.

The results show that the switch from Ethereum 1.0 to Ethereum 2.0 has resulted in a significant decrease in carbon footprint. Ethereum 1.0 had a substantial carbon footprint of 1115.376 KgCO_{2e} before the merge. The carbon footprint, however, has dropped significantly to just 0.274 KgCO_{2e} after Ethereum 2.0 was implemented, representing a remarkable reduction rate of 99.975%.

We compared CO₂ emissions of the smart HYB with different indicators, such as the number of automobile kilometers, to highlight the negative environmental effects of blockchain technology, particularly when it comes to using energy-intensive consensus mechanisms like Proof of Work (Ethereum 1.0). However, when we did the same comparison to the Proof of Stake carbon footprint, this showed Ethereum's commitment to ensuring sustainability and its vision in managing the environmental risks related to blockchain technology.

Our results clearly show that tokenization improves asset sustainability. Our findings corroborate those of previous studies and in particular that of Tawiah et al. [51] who examined the relationship between blockchain technology and the environmental efficiency of 103 US firms from 2015 to 2019, and concluded that the adoption of blockchain technology significantly relates to environmental sustainability.

Tawiah et al. [51] also showed that the relationship between blockchain and sustainability is more pronounced for firms belonging to the finance and technology sectors. These findings support our results about the case study ODDO BHF, which belongs to the financial sector.

Conclusion

This study aims to explore the impact of smart contracts and tokenization, on costs of the trading process and on its carbon footprint. The innovative contribution of the study is that it is the first empirical study to jointly examine cost and environmental dimensions of tokenization, filling a gap in the literature on blockchain applications in the bond market. The comparison before and after Ethereum's transition from Proof-of-Work to Proof-of-Stake provides a unique perspective. From an academic perspective, this study could enhance the understanding of the dual financial environmental trade-offs of digital assets, while from a banking and policy standpoint, it provides practical insights for institutions seeking the integration of financial assets tokenization into their trading strategies and sustainability evaluation of emerging fintech solutions.

As a conclusion, this study highlights the significant cost efficiencies that tokenization can offer to ODDO BHF High Yield Bond. It also underscores the remarkable improvement of the Ethereum blockchain in reducing carbon footprint after transitioning from PoW to PoS, by comparing the carbon footprint of the studied bond before and after "the merge" to PoS.

These findings provide evidence to encourage ODDO BHF to consider implementing blockchain as infrastructure for its bond trading process, as it responds to its interest in new technologies and commitment toward ESG. In addition, it is advisable to use a sustainable blockchain platform, like Ethereum, to guarantee long-term viability as well as avoid reputational risks, including greenwashing.

Although tokenization has improved asset liquidity as well as transaction efficiency and transparency, significant barriers remain such as regulatory ambiguity, technical infrastructure limits, token market volatility, and insufficient public sector engagement [27].

This study bears on the assumption of solid regulation, as it is outside the scope of MiCA and falls under MiFID II. In future research, we aim to focus on studying this type of token within the framework of the DLT Pilot Regime to see how it behaves as a service under MiFID II on new technological infrastructure.

On a broader scale, these results also highlight the need to align with key international regulatory frameworks, including the EBSI (European Blockchain Services Infrastructure) for cross-border blockchain applications, including sustainability tracking and regulatory reporting, and the MiCA (Markets in Crypto-Assets Regulation) for the supervision of cryptocurrencies, promoting market integrity and consumer protection while imposing environmental disclosure requirements for crypto-assets, ensuring accountability in the digital financial ecosystem.

To implement these benchmarks, blockchain-based sustainability solutions must incorporate automated reporting tools that enable real-time measurement and verification of sustainability indicators. This also ensures compliance with regulatory standards (Toucan Protocol, 2023). These tools enable a shift towards a more responsible and climate-conscious future, ensuring that DLT technology becomes a catalyst for global sustainability efforts rather than an obstacle.

This study, additionally, includes several limitations. First, because of the novelty of the subject, there is a lack of datasets providing real historical data on the costs of smart contracts associated with issued smart bonds. Our cost estimates are approximations derived from the Remix programming tool, which are based on a smart contract containing only basic functions. In fact, perhaps availability of more comprehensive datasets for smart bonds could potentially increase accuracy of results on cost reduction. Secondly, our study is based on several assumptions, such as the smart contract being designed for a single investor with a fixed investment amount of 10€ million. These assumptions lead to ignoring the sensitivity of these factors on cost reduction.

We acknowledge, as well, this study does not account for broader market volatility in gas prices over time, nor does it use techniques such as Monte Carlo simulations to model gas-price variability or varying user behaviors. In addition, it does not reflect potential fluctuations in the ETH/EUR exchange rate, as the analysis is based on a single-date estimation. Additionally, the data used in our study was not all extracted on the same date. For instance, all data used in the cost analysis is dated 31/10/2023. However, for carbon calculations, some data is from 2022 and 2021, the most recent published data. The study does not shed light on certain vulnerabilities of tokenization, Ethereum smart contract risks, or potential rebound effects that may arise from Ethereum post-merge.

As future research, we propose that studies should also explore hybrid consensus factors that combine the security of PoW with the efficiency of PoS to optimize sustainability. It is important to assess the long-term security of PoS under high-stakes conditions to address concerns on centralization. Additionally, we aim in the future to conduct further studies focusing on quantifying ESG performance and rating of ODDO BHF in the long-term in case of implementing Ethereum blockchain. Finally, it should be essential to model the economic effects of regulatory frameworks on cryptocurrency markets that would inform policy and innovation strategies.

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Author contributions

Conceptualization: Eya ABID, Methodology: Sina BELKHIRIA and Eya ABID, Data curation: Eya ABID, Investigation: Eya ABID, Sina BELKHIRIA and Wided KHIARI, Validation: Sina BELKHIRIA and Wided KHIARI, Formal analysis: Eya ABID, Supervision: Sina BELKHIRIA and Wided KHIARI, Visualization: Sina BELKHIRIA, Eya ABID and Wided KHIARI, Project administration: Sina BELKHIRIA and Wided KHIARI, Resources: Eya ABID, Sina BELKHIRIA and Wided KHIARI, Writing—original draft: Eya ABID, Writing—review & editing: Sina BELKHIRIA, Wided KHIARI and Eya ABID.

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No datasets were generated or analyzed during the current study.

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